

1. A display system has a resolution of 1280×1024 pixels. If each pixel uses 16-bit color depth (High Color), what is the minimum frame buffer size required?

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|-------------|-------------|
| (A) 1.25 MB | (B) 2.50 MB |
| (C) 5.00 MB | (D) 10.0 MB |

2. Which type of LCD panel offers fast response times ideal for gaming, but suffers from narrow viewing angles and poor color accuracy?

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|-----------------------------|------------------------------|
| (A) TN (Twisted Nematic) | (B) IPS (In-Plane Switching) |
| (C) VA (Vertical Alignment) | (D) OLED |

3. In a CRT or raster-scan display, the movement of the electron beam returning from the end of one scan line to the start of the next scan line is called:

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|--------------------------|----------------------|
| (A) Horizontal Retrace | (B) Vertical Retrace |
| (C) Aspect Ratio Refresh | (D) Raster Blanking |

4. Which printer language is a high-level command language created by Hewlett-Packard for controlling printer features and page formatting?

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|------------------------------------|----------------|
| (A) PCL (Printer Command Language) | (B) PostScript |
| (C) GDI | (D) PDF |

5. What component in an inkjet printer precisely controls the ejection of ink droplets using electrical pulses applied to tiny heating resistors?

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|-------------------------------------|------------------------|
| (A) Thermal Inkjet Printhead Nozzle | (B) Photoreceptor Drum |
| (C) Corona Wire | (D) Fuser Roller |

6. What is the standard native resolution corresponding to 4K Ultra High Definition (UHD)?

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|-------------------------------|-------------------------------|
| (A) 3840×2160 pixels | (B) 2560×1440 pixels |
| (C) 1920×1080 pixels | (D) 4096×2160 pixels |

7. During the laser printing process, which component places a uniform negative electrical charge onto the surface of the photosensitive drum?

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|---|--------------------------|
| (A) Primary Corona Wire / Charging Roller | (B) Laser Diode Assembly |
| (C) Developer Roller | (D) Transfer Roller |

8. Which display connection standard was designed specifically as an open-standard, high-bandwidth replacement for DVI and VGA, featuring a latching connector?

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|-----------------|---------------------|
| (A) DisplayPort | (B) HDMI |
| (C) S-Video | (D) Component Video |

9. In drum plotters, how is vector line drawing along the horizontal and vertical axes achieved?

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|--|---|
| (A) Pen moves horizontally while the drum rotates paper vertically | (B) Paper remains stationary while the pen moves in both X and Y directions |
| (C) Laser beam reflects off mirrors while paper moves on a belt | (D) Thermal pins strike ink ribbons against a rotating cylinder |

10. In digital audio output, if an audio signal is sampled at 44.1 kHz with a bit depth of 16 bits in stereo (2 channels), what is the uncompressed bit rate?

- (A) 1,411.2 kbps
- (B) 705.6 kbps
- (C) 320 kbps
- (D) 128 kbps

11. Which technology dynamic synchronization feature eliminates screen tearing by allowing the GPU to control the monitor refresh rate dynamically?

- (A) G-Sync / FreeSync
- (B) Anti-Aliasing
- (C) Anisotropic Filtering
- (D) Frame Interpolation

12. Which non-impact printer technology uses continuous solid wax blocks that are melted into liquid state before being sprayed onto paper?

- (A) Solid Ink Printer
- (B) Direct Thermal Printer
- (C) Dye-Sublimation Printer
- (D) Dot-Matrix Printer

13. What parameter measures the diagonal distance between pixels of the same color on a display screen?

- (A) Dot Pitch
- (B) Aspect Ratio
- (C) Contrast Ratio
- (D) Viewing Angle

14. What is the function of the Raster Image Processor (RIP) in high-end publishing printers?

- (A) Converts vector graphic descriptions into a high-resolution raster bitmap for printing
- (B) Cleans residual toner from photo drum
- (C) Compresses PDF documents for local disk storage
- (D) Regulates paper tray feed speed

15. What advantage does a 32-bit color depth (True Color with Alpha channel) offer over 24-bit True Color?

- (A) Adds an 8-bit Alpha channel for pixel transparency and blending
- (B) Increases maximum screen refresh rate
- (C) Uses 256 additional primary colors
- (D) Eliminates the requirement for a DAC

16. Which impact printer uses a matrix of small pins (typically 9 or 24 pins) to form characters out of tiny ink dots on paper?

- (A) Dot-Matrix Printer
- (B) Daisy Wheel Printer
- (C) Laser Printer
- (D) Thermal Transfer Printer

17. In projection systems, what is the term for the distortion of an image (making it look like a trapezoid) when projected at an angle onto a flat screen?

- (A) Keystone Distortion
- (B) Chromatic Aberration
- (C) Barrel Distortion
- (D) Parallax Error

18. Which part of an operating system graphics pipeline translates generic OS drawing commands into hardware-specific instructions for the GPU and display device?

- (A) Display Driver
- (B) Frame Buffer
- (C) BIOS interrupt vector
- (D) Spooler Service

19. How many distinct color shades can be displayed on a monochrome display screen with 4-bit pixel depth?

- (A) 16 shades (B) 32 shades
(C) 64 shades (D) 256 shades

20. Which sub-type of LCD technology uses liquid crystal molecules that rotate parallel to the glass substrates when voltage is applied?

- (A) IPS (In-Plane Switching) (B) TN (Twisted Nematic)
(C) CRT (D) STN (Super Twisted Nematic)

21. In laser printers, what is used to discharge specific areas on the photoreceptor drum, creating a latent electrostatic image?

- (A) Focused Laser Light Beam (B) Heated Wire
(C) Magnetic Roller (D) Mechanical Print Pins

22. What output metric measures speaker power output continuously under realistic listening conditions without excessive distortion?

- (A) RMS Power (Root Mean Square) (B) PMPO (Peak Music Power Output)
(C) Impedance (Ohms) (D) Sensitivity (dB)

23. What is the principal cause of burn-in on OLED screens?

- (A) Uneven degradation of organic light-emitting subpixels over long static displays (B) Overheating of liquid crystal fluid
(C) High voltage arc in cathode ray tube (D) Exhaustion of toner powder

24. Which printing technology utilizes microscopic heating elements to vaporize dye from a ribbon directly into a gaseous state, diffusing onto photo paper?

- (A) Dye-Sublimation Printer (B) Thermal Transfer Printer
(C) Inkjet Printer (D) Laser Printer

25. In surround sound audio specifications (e.g., 7.1 surround sound), what does the number after the decimal point (.1) specify?

- (A) Single dedicated Low-Frequency Effects (LFE) subwoofer channel (B) Single center dialog speaker
(C) One overhead height speaker (D) Mono rear speaker

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)